

```

In [ ]: from abc import ABC, abstractmethod
        from dataclasses import dataclass
        from typing import Annotated

        import random

        import numpy as np
        from numpy.lib.stride_tricks import as_strided

        from geneticengine.grammar.metahandlers.ints import IntRange
        from geneticengine.grammar import extract_grammar
        from geneticengine.problems import SingleObjectiveProblem, MultiObjectiveProblem
        from geneticengine.random.sources import NativeRandomSource
        from geneticengine.algorithms.gp.gp import GeneticProgramming
        from geneticengine.evaluation.budget import TimeBudget, EvaluationBudget
        from geneticengine.representations.tree.initializations import MaxDepthDecider, Full
        from geneticengine.representations.tree.operators import GrowInitializer, PositionI
        from geneticengine.algorithms.gp.operators.initializers import HalfAndHalfInitializ
        from geneticengine.representations.tree.treebased import TreeBasedRepresentation
        from geneticengine.representations.grammatical_evolution.structured_ge import Struc
        from geneticengine.evaluation.recorder import CSVSearchRecorder
        from geneticengine.evaluation.tracker import ProgressTracker
        from geneticengine.evaluation.parallel import ParallelEvaluator

        from geneticengine.algorithms.gp.operators.combinators import ParallelStep, Sequenc
        from geneticengine.algorithms.gp.operators.crossover import GenericCrossoverStep
        from geneticengine.algorithms.gp.operators.elitism import ElitismStep
        from geneticengine.algorithms.gp.operators.mutation import GenericMutationStep
        from geneticengine.algorithms.gp.operators.novelty import NoveltyStep
        from geneticengine.algorithms.gp.operators.selection import LexicaseSelection, Tour

        from sklearn.datasets import load_breast_cancer

        import pandas as pd

        from sklearn.tree import DecisionTreeClassifier
        from sklearn.metrics import f1_score, confusion_matrix, classification_report, Conf
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler, OneHotEncoder
        from sklearn.ensemble import RandomForestClassifier
        from sklearn.linear_model import LogisticRegression

        import time

        import seaborn as sns
        import matplotlib.pyplot as plt

```

```

In [22]: def plot_roc(fpr, tpr):
          plt.plot(fpr, tpr, label='ROC curve')
          plt.xlabel('False Positive Rate')
          plt.ylabel('True Positive Rate')
          plt.title('ROC curve')
          plt.legend()
          plt.show()

```

```
def evaluate(predictions, FIXED_FPR = 0.05):
    fprs, tprs, thresholds = roc_curve(y_test, predictions)
    plot_roc(fprs, tprs)
    tpr = tprs[fprs<FIXED_FPR][-1]
    fpr = fprs[fprs<FIXED_FPR][-1]
    threshold = thresholds[fprs<FIXED_FPR][-1]

    print("AUC:", roc_auc_score(y_test, predictions))
    to_pct = lambda x: str(round(x, 4) * 100) + "%"
    print("TPR: ", to_pct(tpr), "\nFPR: ", to_pct(fpr), "\nThreshold: ", round(thre
```

In [23]: df = pd.read_csv('base.csv')

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   fraud_bool                            1000000 non-null  int64
1   income                                1000000 non-null  float64
2   name_email_similarity                 1000000 non-null  float64
3   prev_address_months_count            1000000 non-null  int64
4   current_address_months_count         1000000 non-null  int64
5   customer_age                         1000000 non-null  int64
6   days_since_request                   1000000 non-null  float64
7   intended_balcon_amount                1000000 non-null  float64
8   payment_type                          1000000 non-null  object
9   zip_count_4w                         1000000 non-null  int64
10  velocity_6h                          1000000 non-null  float64
11  velocity_24h                         1000000 non-null  float64
12  velocity_4w                          1000000 non-null  float64
13  bank_branch_count_8w                 1000000 non-null  int64
14  date_of_birth_distinct_emails_4w     1000000 non-null  int64
15  employment_status                    1000000 non-null  object
16  credit_risk_score                    1000000 non-null  int64
17  email_is_free                        1000000 non-null  int64
18  housing_status                       1000000 non-null  object
19  phone_home_valid                     1000000 non-null  int64
20  phone_mobile_valid                   1000000 non-null  int64
21  bank_months_count                    1000000 non-null  int64
22  has_other_cards                      1000000 non-null  int64
23  proposed_credit_limit                1000000 non-null  float64
24  foreign_request                      1000000 non-null  int64
25  source                               1000000 non-null  object
26  session_length_in_minutes            1000000 non-null  float64
27  device_os                            1000000 non-null  object
28  keep_alive_session                   1000000 non-null  int64
29  device_distinct_emails_8w            1000000 non-null  int64
30  device_fraud_count                   1000000 non-null  int64
31  month                               1000000 non-null  int64
dtypes: float64(9), int64(18), object(5)
memory usage: 244.1+ MB
```

```
In [24]: df.head(2)
```

```
Out[24]:
```

	fraud_bool	income	name_email_similarity	prev_address_months_count	current_address_
0	0	0.3	0.986506		-1
1	0	0.8	0.617426		-1

2 rows × 32 columns

```
In [25]: categorical_features = [col for col in df.columns if df[col].dtype == 'object']

encoder = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

encoded_data = encoder.fit_transform(df[categorical_features])

encoded_df = pd.DataFrame(encoded_data, columns=encoder.get_feature_names_out(categ

df = df.drop(columns=categorical_features).reset_index(drop=True)
df = pd.concat([df, encoded_df], axis=1)

df.head(2)
```

```
Out[25]:
```

	fraud_bool	income	name_email_similarity	prev_address_months_count	current_address_
0	0	0.3	0.986506		-1
1	0	0.8	0.617426		-1

2 rows × 53 columns

```
In [26]: df['month'].value_counts().sort_index()
```

```
Out[26]: month
0      132440
1      127620
2      136979
3      150936
4      127691
5      119323
6      108168
7       96843
Name: count, dtype: int64
```

```
In [27]: X = df.drop(['fraud_bool'], axis=1)
y = df['fraud_bool']

X_train = X[X['month'] < 6]
X_test = X[X['month'] >= 6]
y_train = y[X['month'] < 6]
y_test = y[X['month'] >= 6]
```

```
X_train.drop('month', axis=1, inplace=True)
X_test.drop('month', axis=1, inplace=True)
```

/tmp/ipykernel_14963/593396559.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
X_train.drop('month', axis=1, inplace=True)
```

/tmp/ipykernel_14963/593396559.py:10: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
X_test.drop('month', axis=1, inplace=True)
```

```
In [28]: feature_names = X_train.columns.tolist()
         n_features = len(feature_names)
```

```
In [29]: print(y_train.value_counts(), y_test.value_counts())
```

```
fraud_bool
0      786838
1       8151
Name: count, dtype: int64
fraud_bool
0      202133
1       2878
Name: count, dtype: int64
```

```
In [30]: features_to_scale = [col for col in X_train.select_dtypes(include=np.number).columns
                             if col != 'fraud_bool']

scaler = StandardScaler()
X_train[features_to_scale] = scaler.fit_transform(X_train[features_to_scale])
# X_val[features_to_scale] = scaler.transform(X_val[features_to_scale])
X_test[features_to_scale] = scaler.transform(X_test[features_to_scale])
```

/tmp/ipykernel_14963/3223336677.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
X_train[features_to_scale] = scaler.fit_transform(X_train[features_to_scale])
```

/tmp/ipykernel_14963/3223336677.py:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
X_test[features_to_scale] = scaler.transform(X_test[features_to_scale])
```

```
In [31]: baseline_model = LogisticRegression(random_state=42, class_weight='balanced')
         # baseline_model = RandomForestClassifier(random_state=42, class_weight='balanced')

         baseline_model.fit(X_train, y_train)
```

```

baseline_y_pred = baseline_model.predict(X_test)

# baseline_f1 = f1_score(y_test, baseline_y_pred)
predictions = baseline_model.predict_proba(X_test)[:, 1]

fpr, tpr, _ = roc_curve(y_test, predictions)

target_fpr = 0.05
baseline_recall_at_fpr = 0.0

if np.any(fpr <= target_fpr):
    valid_indices = np.where(fpr <= target_fpr)[0] #where returns a tuple
    best_index = valid_indices[np.argmax(tpr[valid_indices])]
    baseline_tpr_at_fpr = tpr[best_index]

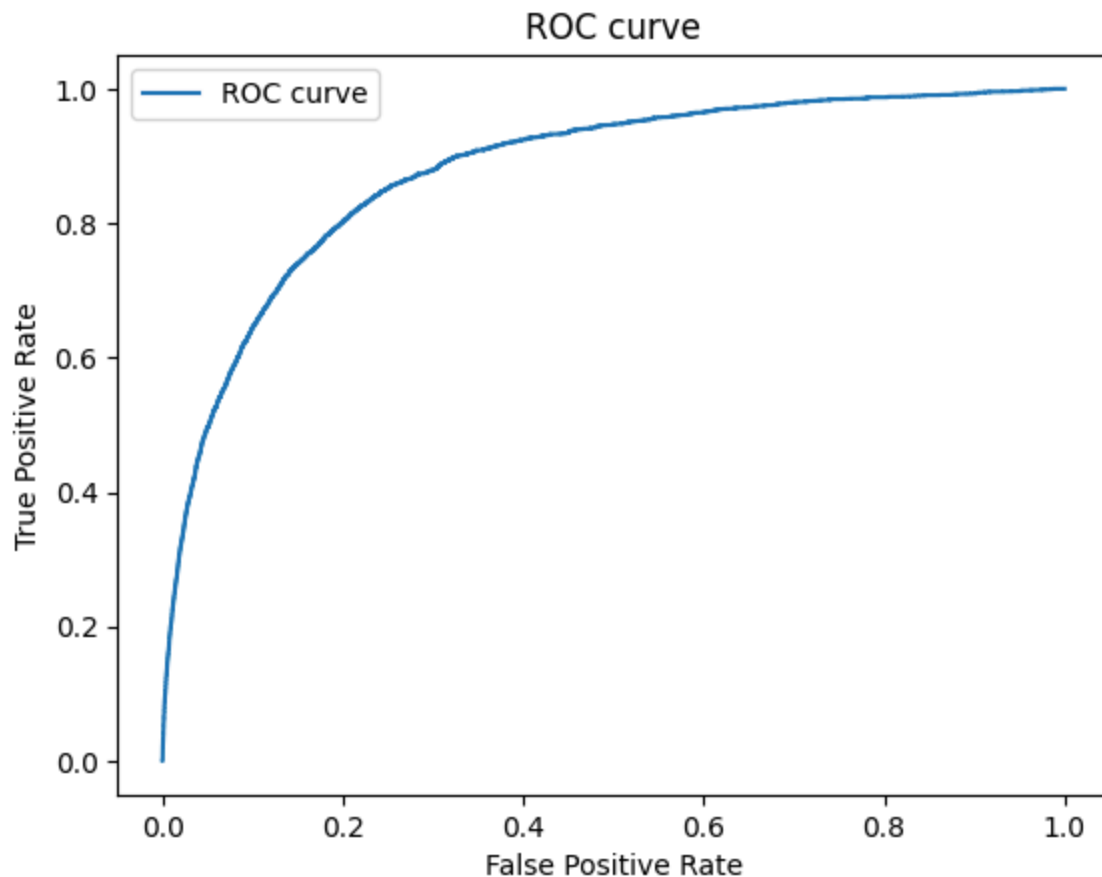
print(f"TPR: {baseline_tpr_at_fpr}")

evaluate(predictions)

print(classification_report(y_test, baseline_y_pred))
cm_baseline = confusion_matrix(y_test, baseline_y_pred)
disp_baseline = ConfusionMatrixDisplay(confusion_matrix=cm_baseline, display_labels
disp_baseline.plot(cmap=plt.cm.Blues)
plt.show()

```

TPR: 0.49687282835302293



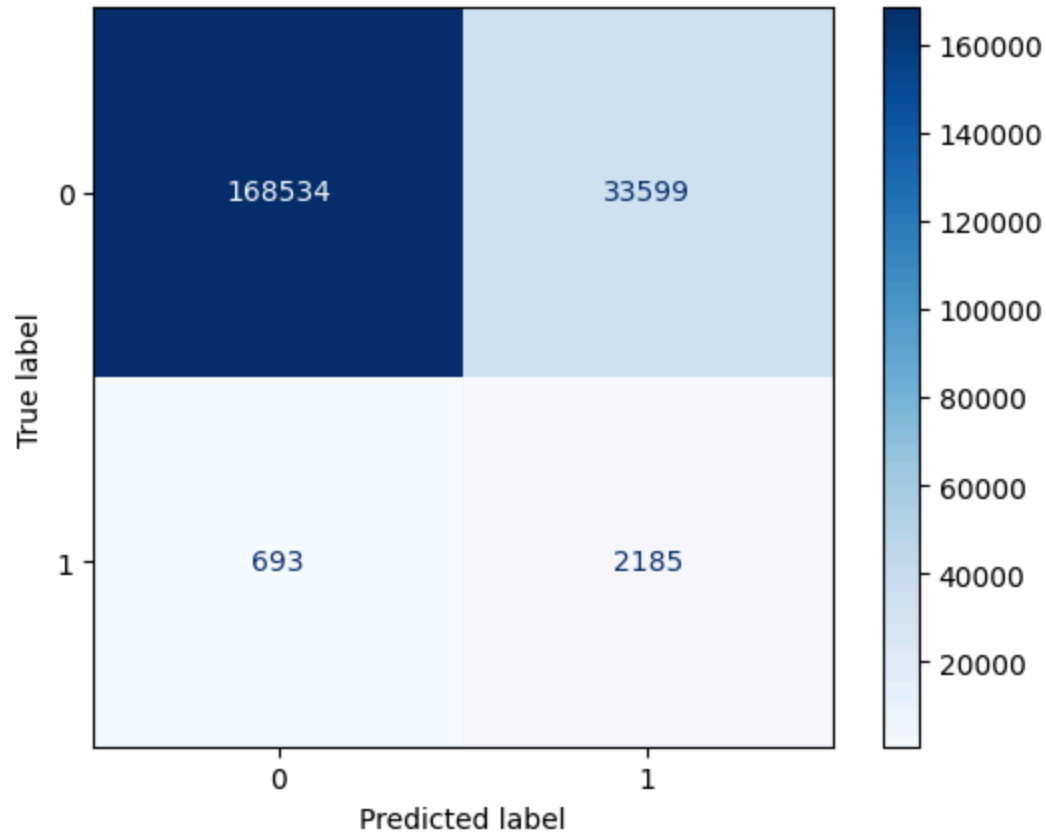
AUC: 0.8778659233740538

TPR: 49.69%

FPR: 4.99%

Threshold: 0.76

	precision	recall	f1-score	support
0	1.00	0.83	0.91	202133
1	0.06	0.76	0.11	2878
accuracy			0.83	205011
macro avg	0.53	0.80	0.51	205011
weighted avg	0.98	0.83	0.90	205011



```
In [32]: class Value(ABC):
        def evaluate(self):
            pass

        class Scalar(ABC):
            pass

        class Vectorial(ABC):
            pass
```

```
In [33]: @dataclass
        class PastValues(Vectorial):
            index: Annotated[int, IntRange(0, n_features-1)]
            lookback: Annotated[int, IntRange(5, 15)]

            # def evaluate(self, X_np, i):
```

```

#     start_index = max(0, i-self.lookback+1)
#     values = X_np[start_index: i+1, self.index]
#     if len(values) < self.lookback: #if there are not enough past values
#         padding = np.zeros(self.lookback - len(values))
#         values = np.concatenate((padding, values))
#     return values

def evaluate(self, X_np):
    feature_col = X_np[:, self.index]
    padded_col = np.concatenate([np.zeros(self.lookback-1), feature_col])

    shape = (feature_col.shape[0], self.lookback)
    strides = (padded_col.strides[0], padded_col.strides[0])
    rolling_windows = as_strided(padded_col, shape=shape, strides=strides)

    return rolling_windows

def __str__(self):
    return f"past_values({feature_names[self.index]}, {self.lookback})"

```

```

In [34]: @dataclass
class Mean(Scalar):
    arr: Vectorial
    def evaluate(self, X_np):
        v = self.arr.evaluate(X_np)
        return np.mean(v)
    def __str__(self):
        return f"mean({self.arr})"

@dataclass
class Max(Scalar):
    arr: Vectorial
    def evaluate(self, X_np):
        v = self.arr.evaluate(X_np)
        return np.max(v)
    def __str__(self):
        return f"max({self.arr})"

@dataclass
class Min(Scalar):
    arr: Vectorial
    def evaluate(self, X_np):
        v = self.arr.evaluate(X_np)
        return np.min(v)
    def __str__(self):
        return f"min({self.arr})"

@dataclass
class Std(Scalar):
    arr: Vectorial
    def evaluate(self, X_np):
        v = self.arr.evaluate(X_np)
        return np.std(v)
    def __str__(self):
        return f"std({self.arr})"

```

```
In [35]: @dataclass #Scalar Features (1)
class ScalarVar(Scalar):
    index: Annotated[int, IntRange(0,n_features-1)]

    def evaluate(self, X_np):
        return X_np[:, self.index]

    def __str__(self):
        return feature_names[self.index]

# @dataclass #Vectorial Features ([1,2,3,4])
# class VectorialVar(Vectorial):
#     index: Annotated[int, IntRange(1,2)]

#     def evaluate(self, X):
#         return X[self.index]
```

```
In [36]: #scalar -> scalar
@dataclass
class Add(Scalar):
    right: Scalar
    left: Scalar

    def evaluate(self, X_np):
        return self.left.evaluate(X_np) + self.right.evaluate(X_np)

    def __str__(self):
        return f"({self.left} + {self.right})"

@dataclass
class Subtract(Scalar):
    right: Scalar
    left: Scalar

    def evaluate(self, X_np):
        return (self.left.evaluate(X_np)) - (self.right.evaluate(X_np))

    def __str__(self):
        return f"({self.left} - {self.right})"

@dataclass
class Multiply(Scalar):
    right: Scalar
    left: Scalar

    def evaluate(self, X_np):
        return self.left.evaluate(X_np) * self.right.evaluate(X_np)

    def __str__(self):
        return f"({self.left} * {self.right})"

@dataclass
class SafeDiv(Scalar):
    right: Scalar
    left: Scalar
```



```

def evaluate(self, X_np):
    right_val = self.right.evaluate(X_np)
    if right_val==0:
        return 0.0
    left_val = self.left.evaluate(X_np)
    return left_val/right_val

def __str__(self):
    return f"SafeDiv({self.left}/{self.right})"

```

```

In [37]: grammar = extract_grammar([PastValues, Mean, Max, Min, Std, ScalarVar, Add, Subtrac
# grammar = extract_grammar([Add, Subtract, Multiply, ScalarVar], Scalar)
print(f"Grammar: {repr(grammar)}")

```

```

Grammar: Grammar<Starting=Scalar,Productions={
Scalar -> Mean(arr: Vectorial)|
        Max(arr: Vectorial)|
        Min(arr: Vectorial)|
        Std(arr: Vectorial)|
        ScalarVar(index: Annotated[int])|
        Add(right: Scalar, left: Scalar)|
        Subtract(right: Scalar, left: Scalar)|
        Multiply(right: Scalar, left: Scalar)

```

```

Vectorial -> PastValues(index: Annotated[int], lookback: Annotated[int])
}

```

```

In [38]: X_train_np = X_train.to_numpy()
X_test_np = X_test.to_numpy()

def fitness_function(individual: Value): #individual -> expression
    start = time.perf_counter()

    train_feature = individual.evaluate(X_train_np)
    test_feature = individual.evaluate(X_test_np)

    if train_feature.ndim == 0:
        train_feature = np.full(X_train_np.shape[0], train_feature)
    if test_feature.ndim == 0:
        test_feature = np.full(X_test_np.shape[0], test_feature)

    X_train_augmented = np.c_[X_train_np, np.array(train_feature).reshape(-1,1)]
    X_test_augmented = np.c_[X_test_np, np.array(test_feature).reshape(-1,1)]

    model = LogisticRegression(random_state=42, class_weight='balanced', max_iter=1
model.fit(X_train_augmented, y_train)

    y_probs = model.predict_proba(X_test_augmented)[: , 1]

    fpr, tpr, thresholds = roc_curve(y_test, y_probs)

    target_fpr = 0.05
    recall_at_fpr = 0.0

    if np.any(fpr <= target_fpr):

```

```

        valid_indices = np.where(fpr<=target_fpr)[0]
        best_indice = valid_indices[np.argmax(tpr[valid_indices])]
        tpr_at_fpr = tpr[best_indice]

    roc_auc = roc_auc_score(y_test, y_probs)

    # tpr_diff = tpr_at_fpr-baseline_recall_at_fpr

    features, num_operations = analyse_complexity(individual)

    # num_features = len(features)

    if num_operations == 0:
        num_operations = 1

    elapsed = time.perf_counter() - start

    return [float(tpr_at_fpr), float(roc_auc), num_operations, float(elapsed)]

def analyse_complexity(individual: Value):
    if isinstance(individual, ScalarVar):
        return {individual.index}, 0 #unique feature
    if isinstance(individual, PastValues):
        return {individual.index}, 0

    total_features = set()
    total_operations = 1
    if hasattr(individual, 'left') and hasattr(individual, 'right'):
        left_features, left_operations = analyse_complexity(individual.left)
        right_features, right_operations = analyse_complexity(individual.right)
        total_features.update(left_features)
        total_features.update(right_features)
        total_operations += left_operations + right_operations
    elif hasattr(individual, 'arr'):
        # if isinstance(individual, Mean) and isinstance(individual.arr, PastValues)
        # total_operations += individual.arr.lookback
        arr_features, arr_operations = analyse_complexity(individual.arr)
        total_features.update(arr_features)
        total_operations += arr_operations
    return total_features, total_operations

```

```

In [39]: def lexicase_step():
    return ParallelStep(
        [
            ElitismStep(),
            NoveltyStep(),
            SequenceStep(
                LexicaseSelection(True),
                # TournamentSelection(5),
                GenericCrossoverStep(0.9),
                GenericMutationStep(0.1),
            ),
        ],
        weights=[.05,.05,.8]
        # weights=[.9]
    )

```

```

prob = MultiObjectiveProblem(
    fitness_function=fitness_function,
    minimize=[False, False, True, True],
)
r = NativeRandomSource(123)
alg = GeneticProgramming(
    problem=prob,
    budget=TimeBudget(600),
    population_size=50,
    representation=TreeBasedRepresentation(grammar, FullDecider(r, grammar, 4)),
    random=r,
    step=lexicase_step(),
    tracker=ProgressTracker(
        prob,
        recorders=[CSVSearchRecorder(
            csv_path='feedzai_output.csv',
            problem=prob,
            fields={
                "Eval Time": lambda t,i,p: i.get_fitness(p).fitness_components[
                "TPR": lambda t,i,p: i.get_fitness(p).fitness_components[0],
                'AUC': lambda t,i,p: i.get_fitness(p).fitness_components[1],
                'TPR Baseline Difference': lambda t,i,p: i.get_fitness(p).fitne
                "Expression": lambda t, i, p: i.get_phenotype(),
                'Num Operations': lambda t,i,p: i.get_fitness(p).fitness_compon
            },
            only_record_best_individuals=False)]
    )
)

solutions = alg.search()

```

In [40]: `sorted_solutions = sorted(solutions, key=lambda ind: ind.get_fitness(prob).fitness_`

```

unique_solutions = []
seen_expressions = set()
for s in sorted_solutions:
    expr = str(s.get_phenotype())
    if expr not in seen_expressions:
        unique_solutions.append(s)
        seen_expressions.add(expr)

```

In [41]: `for i,s in enumerate(unique_solutions[:10]):`
`print(f"Solution {i}:")`
`print(f"Expression: {s.get_phenotype()}")`
`print(f"Fitness: {s.get_fitness(prob).fitness_components}\n")`

Solution 0:
 Expression: ((employment_status_CC * income) - min(past_values(payment_type_AB, 1 2)))
 Fitness: [0.4979152189020153, 0.8779001380437468, 3, 5.451892750999832]

Solution 1:
 Expression: ((housing_status_BA * velocity_24h) + std(past_values(housing_status_BF, 10)))
 Fitness: [0.4979152189020153, 0.8779917152299015, 3, 4.519290186999569]

Solution 2:
 Expression: ((current_address_months_count - payment_type_AA) * (payment_type_AC + email_is_free))
 Fitness: [0.4975677553856845, 0.8779512297043484, 3, 4.575065306000397]

Solution 3:
 Expression: ((session_length_in_minutes * device_os_windows) - std(past_values(phone_mobile_valid, 12)))
 Fitness: [0.49722029186935374, 0.8778799021569087, 3, 4.643571350999991]

Solution 4:
 Expression: (mean(past_values(device_os_other, 13)) * (device_os_macintosh * device_distinct_emails_8w))
 Fitness: [0.49722029186935374, 0.8778596387663168, 3, 4.423816976000126]

Solution 5:
 Expression: ((customer_age * source_INTERNET) * min(past_values(zip_count_4w, 8)))
 Fitness: [0.49687282835302293, 0.8778698632867818, 3, 5.369051586999376]

Solution 6:
 Expression: (std(past_values(foreign_request, 14)) + min(past_values(employment_status_CB, 5)))
 Fitness: [0.49687282835302293, 0.8778716733775769, 3, 4.991269579999425]

Solution 7:
 Expression: ((source_INTERNET * intended_balcon_amount) * min(past_values(source_INTERNET, 11)))
 Fitness: [0.49687282835302293, 0.8778659233740538, 3, 4.152046834999965]

Solution 8:
 Expression: (std(past_values(housing_status_BF, 13)) * (phone_home_valid * current_address_months_count))
 Fitness: [0.49687282835302293, 0.8781293285429176, 3, 6.903377322000779]

Solution 9:
 Expression: (mean(past_values(zip_count_4w, 7)) * (velocity_4w - date_of_birth_distinct_emails_4w))
 Fitness: [0.49687282835302293, 0.8778659233740538, 3, 4.721473571999923]

```
In [58]: top_n_solutions = unique_solutions[:3]
```

```
In [59]: X_train_np = X_train.to_numpy()
X_test_np = X_test.to_numpy()
X_train_fe = X_train.copy()
X_test_fe = X_test.copy()
```

```

for i, target in enumerate(top_n_solutions):
    expression = target.get_phenotype()
    # train_feature_values = [expression.evaluate(X_train_np, i) for i in range(len(X_train_np))]
    # test_features_values = [expression.evaluate(X_test_np, i) for i in range(len(X_test_np))]

    train_features_values = expression.evaluate(X_train_np)
    test_features_values = expression.evaluate(X_test_np)

    X_train_fe[f'new_feature_{i}'] = train_features_values
    X_test_fe[f'new_feature_{i}'] = test_features_values

```

In [60]: X_train_fe.head(5)

Out[60]:

	income	name_email_similarity	prev_address_months_count	current_address_months_count
0	-0.847913	1.676177	-0.395497	-0.7021
1	0.866055	0.407216	-0.395497	0.009
2	0.866055	1.711249	-0.167211	-0.824
3	0.180468	-0.082125	-0.121554	-0.824
4	1.208849	1.180395	-0.395497	-0.657

5 rows × 54 columns

In [61]:

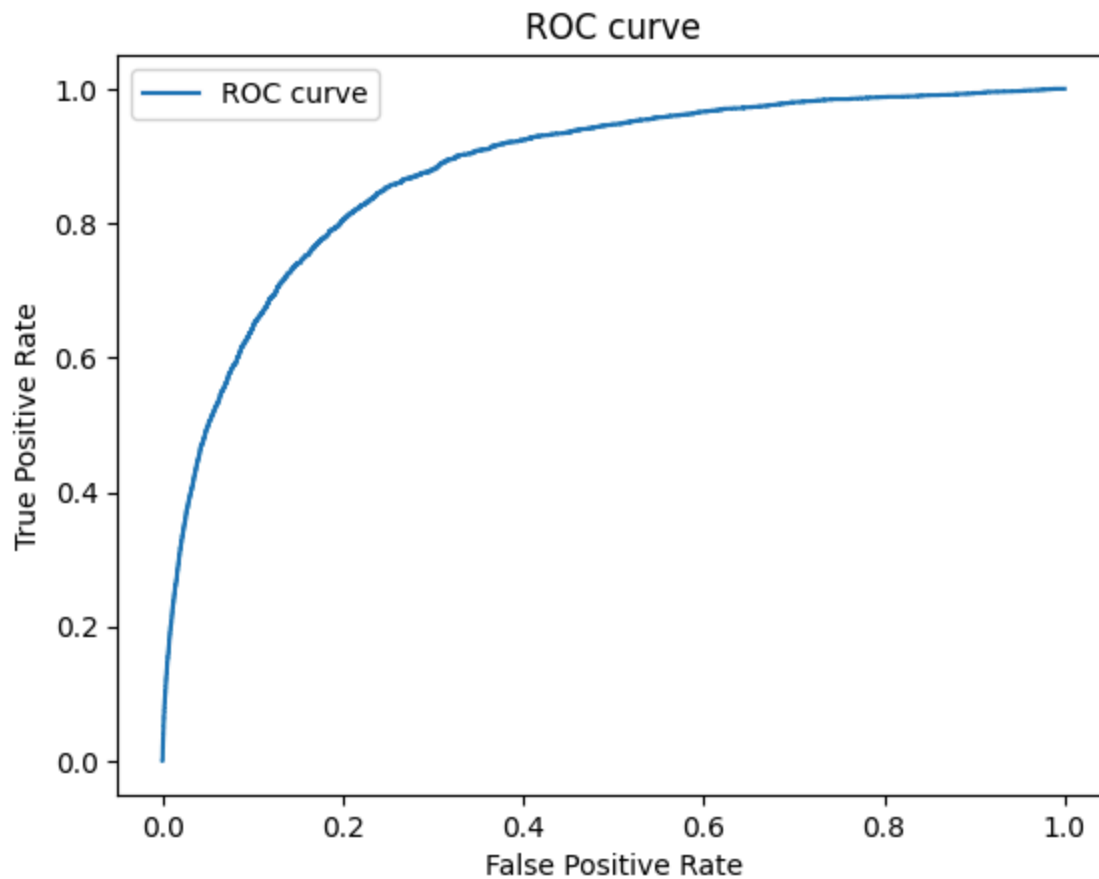
```

model_with_new_features = LogisticRegression(random_state=42, class_weight='balance')
# model_with_new_features = RandomForestClassifier(random_state=42, class_weight='balance')
model_with_new_features.fit(X_train_fe, y_train)

y_pred_with_new_features = model_with_new_features.predict(X_test_fe)

predictions_with_new_features = model_with_new_features.predict_proba(X_test_fe)[:, 1]
evaluate(predictions_with_new_features)

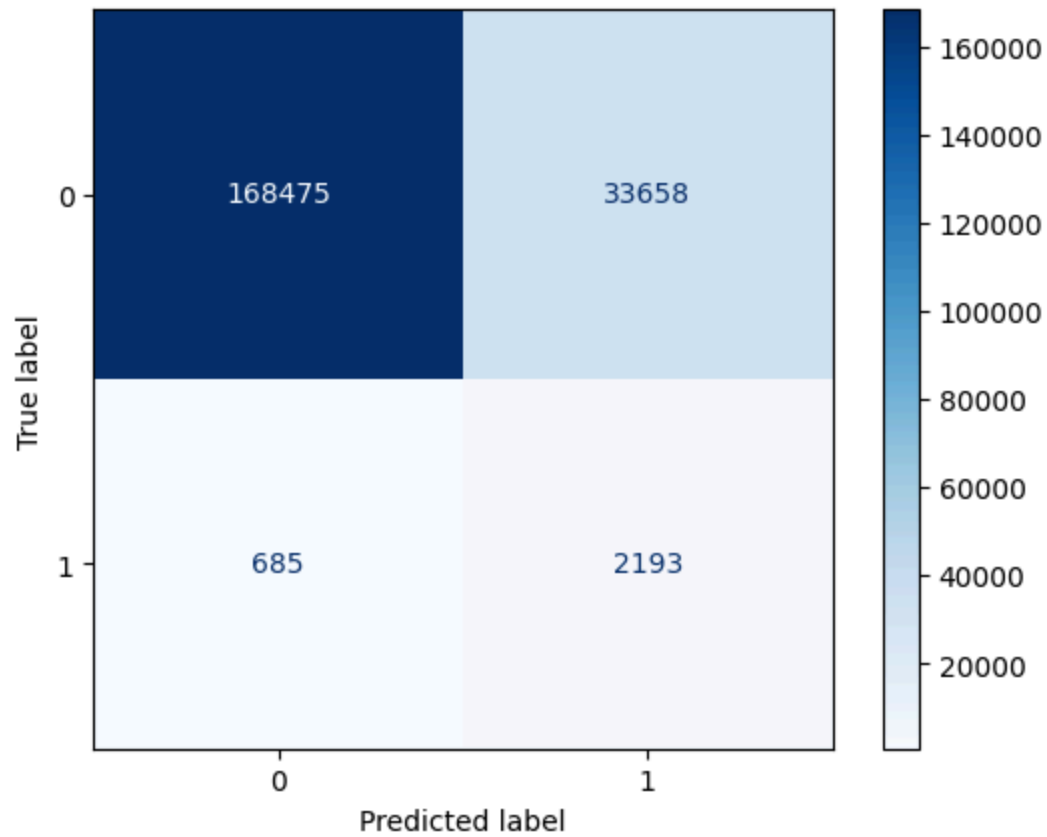
```



AUC: 0.8780600018935647
 TPR: 49.65%
 FPR: 4.979999999999995%
 Threshold: 0.77

```
In [62]: print(classification_report(y_test, y_pred_with_new_features))
cm = confusion_matrix(y_test, y_pred_with_new_features)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=model_with_new_fe
disp.plot(cmap=plt.cm.Blues)
plt.show()
```

	precision	recall	f1-score	support
0	1.00	0.83	0.91	202133
1	0.06	0.76	0.11	2878
accuracy			0.83	205011
macro avg	0.53	0.80	0.51	205011
weighted avg	0.98	0.83	0.90	205011



```
In [63]: temp_df = X_train_fe.copy()
temp_df['fraud_bool'] = y_train
```

```
In [64]: correlation_matrix = temp_df.corr()
plt.figure(figsize=(40,40))
sns.heatmap(correlation_matrix, annot=True)
plt.show()
```

